LaRT and RT-scat: Clumpy and Rotating Lyα Models of Kym et al. (2026)

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1. Purpose

Kym, Chang, Gronke & Seon (2026, arXiv:2609.08396) computed Lyα spectra of rotating smooth and clumpy spheres with the code RT-scat (S.-J. Chang). While recomputing their Figure 9 with LaRT_v2.00, the two codes were compared directly on a static clumpy sphere, and the smooth rotating sphere was examined for numerical artifacts. This note records the setup, the measured results, the problems found in RT-scat, and the changes made to LaRT as a consequence. All numbers below were measured in these runs; the inputs, outputs and analysis notebooks are kept in `combine/Models_for_HGKym/` (`static_fc10/`, `diag_peel_asym/`, `diag_beta899/`).

The RT-scat source used is `SJChang_codes/code_clumpy_rotating` (clumpy) and `code_smooth_rotating_241201` (smooth). Its run parameters are set by editing the source before each compilation, so the values found in the source are not necessarily those used for the paper; only the algorithms are taken from it.

2. Static clumpy sphere: setup

Both codes were run for the same model: a sphere of radius R_o filled with clumps of radius $r_{\rm cl} = 10^{-3} R_o$ at covering factor $f_c = 10$ and mean radial column $N_{\rm HI} = 10^{18}\,{\rm cm}^{-2}$ (clump center-to-surface column $N_{\rm HI,cl} = \frac{3}{4} N_{\rm HI}/f_c = 7.5 \times 10^{16}\,{\rm cm}^{-2}$), $T = 10^4$ K, no dust, no bulk motion, the H I Lyα fine-structure doublet, a central point source emitting a Gaussian of

$\sigma = 1\,\mathrm{km\,s^{-1}}$ around each doublet member, and 10^6 photon packets. Without rotation the model is statistically spherically symmetric, so every viewing angle must give the same spectrum apart from the scatter of one clump realization. Two estimators were used in each code: the image-summed peel-off spectrum of observers at $\theta = 0.1°, 15°, \ldots, 75°, 89.9°$ (azimuth $\phi = 60°$, the RT-scat observer set), and escaping photons binned in $\mu = \cos\theta$ over all azimuths (40 bins in $[-1, 1]$).

	RT-scat	LaRT_v2.00
clump centers	within $\pm(\Delta x/2 - r_{\rm cl})$ of a cell center	uniform over the sphere
clumps at $r < 0.1R_o$	none	yes
overlap	allowed	not allowed (random sequential adsorption)
clump number	$(4/3)f_c(R_o/r_{\rm cl})^2 \times (1-0.1^3)$	$(4/3)f_c(R_o/r_{\rm cl})^2$
peel-off τ cutoff	300	745.2, now 300 (Section 5.)

Section 3.2 of the paper states that the clumps are uniformly distributed within the sphere and do not overlap, which is what LaRT does.

3. Static clumpy sphere: results

Table 1 and Figure 1 summarize the angle dependence. LaRT is isotropic in both estimators, and so are the RT-scat μ bins. The RT-scat peel-off spectra at $\theta = 0.1°$ and 89.9° are not: they carry 24% and 12% more flux than the mean and a filled-in line center, while 15°–75° agree with one another.

Table 1: Angle dependence of the static $f_c = 10$ spectra: integrated flux relative to the mean over the seven angles, and $\sum|S - \langle S\rangle| / \sum\langle S\rangle$ of the unit-area spectra.

estimator	flux / mean	shape deviation	peaks [km s^{-1}]
LaRT peel-off	0.991–1.010	0.1–0.6%	−43, +47
LaRT μ bins	0.991–1.005	0.6–1.5%	−43, +47
RT-scat μ bins	0.980–1.032	1.0–2.6%	−43, +42
RT-scat peel-off, 15°–75°	0.928–0.932	6.4–6.9%	−43, +47
RT-scat peel-off, 0.1°	1.238	22.8%	−33, +37
RT-scat peel-off, 89.9°	1.116	10.4%	−38, +42

The angle-averaged spectra of the two codes agree to 2.7% (Figure 2), but the $\theta = 89.9°$ peel-off spectra differ by 20%.

Scatterings and run time. LaRT scattered 5.92×10^4 times per photon, RT-scat 2.85×10^4 times (2.849×10^4 and 2.843×10^4 for photons born in the two doublet members). RT-scat ran 45.5 min on 144 ranks of lart2+lart1 (Xeon Gold 6154); LaRT ran 76.3 min with the peel-off cutoff at $\tau = 745.2$ and 42.2 min with it at $\tau = 300$, both on 144 ranks of lart4+lart3 (Xeon Gold 6254). Per scattering, LaRT took 11.1 μs (745.2) and 6.2 μs (300) against 13.8 μs for RT-scat (on different processors). The factor of two in the scattering number is not explained yet; candidates are the empty center and the clump-free sheets of RT-scat (Section 4.1).

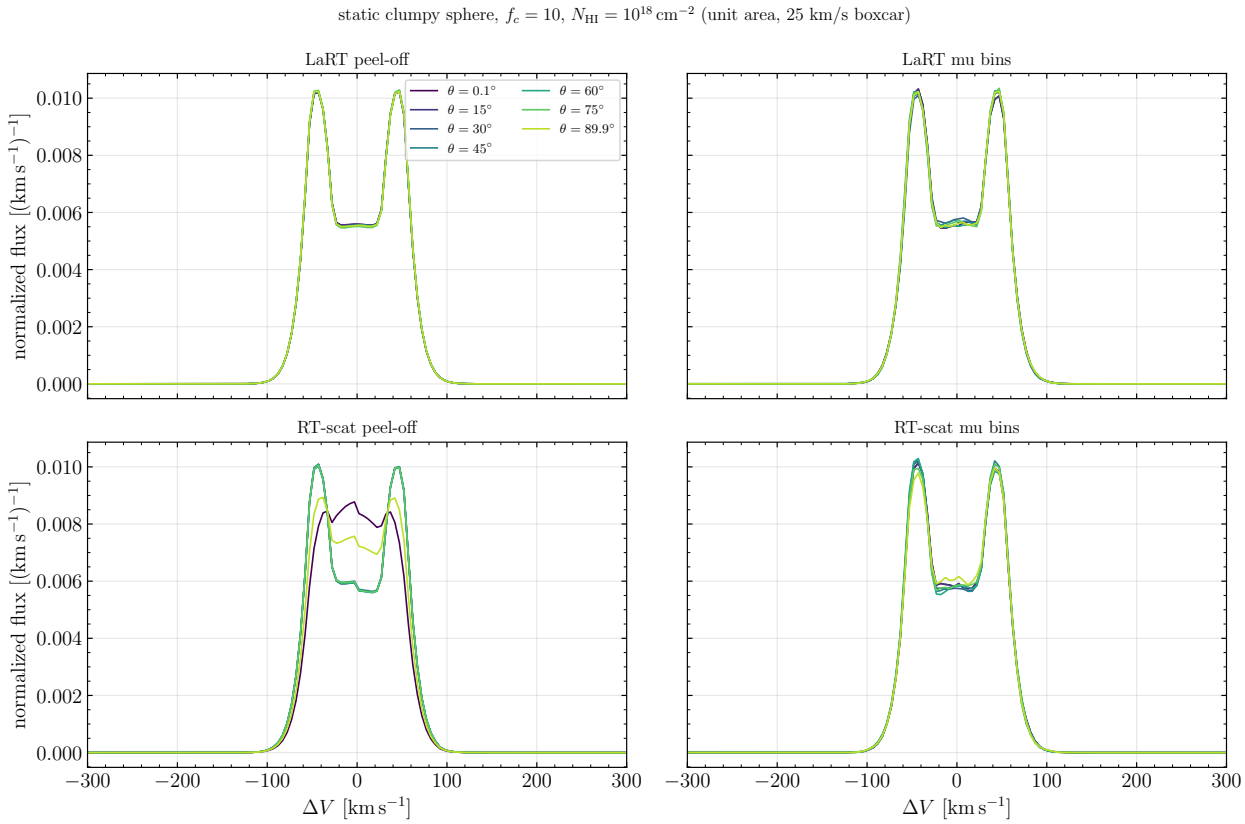


Figure 1: Spectra at seven viewing angles, normalized to unit area and smoothed with a 25 km s^{-1} boxcar. Top: LaRT; bottom: RT-scat; left: peel-off; right: μ bins.

4. Problems found in RT-scat

4.1 Clump-free sheets along the cell faces

RT-scat assigns each clump to a cell of its 201^3 Cartesian grid and draws its center within $\pm(\Delta x/2 - r_{\rm cl})$ of the cell center (`set_clumps_mod.f90`), so no clump crosses a cell face. Thin sheets of width $\sim r_{\rm cl}$ around the planes $x, y, z = $ const are therefore nearly free of clumps. Lines of sight parallel to those sheets ($\theta \approx 0°$: the x and y planes; $\theta \approx 90°$: the z planes) leak more line-center light, which is what the RT-scat peel-off spectra at $0.1°$ and $89.9°$ show. The μ bins average over all azimuths and a range of θ, which dilutes the effect.

The placement was reproduced in a box of 24^3 cells (about 4.3×10^4 clumps, `static_fc10/rtscat_clump_channels.py`); the fraction of rays of length $0.12R_o$ ($\phi = 60°$) that meet no clump is

placement	$\theta = 0.1°$	$45°$	$89.9°$
RT-scat (confined to cells)	0.390	0.281	0.329
uniform (same number)	0.301	0.296	0.293

The clumpy spectra of the paper (Figure 9) are the peel-off spectra of the $\theta = 89.9°$ observer, so they probably contain this leak; this was not checked against the RT-scat outputs of the paper.

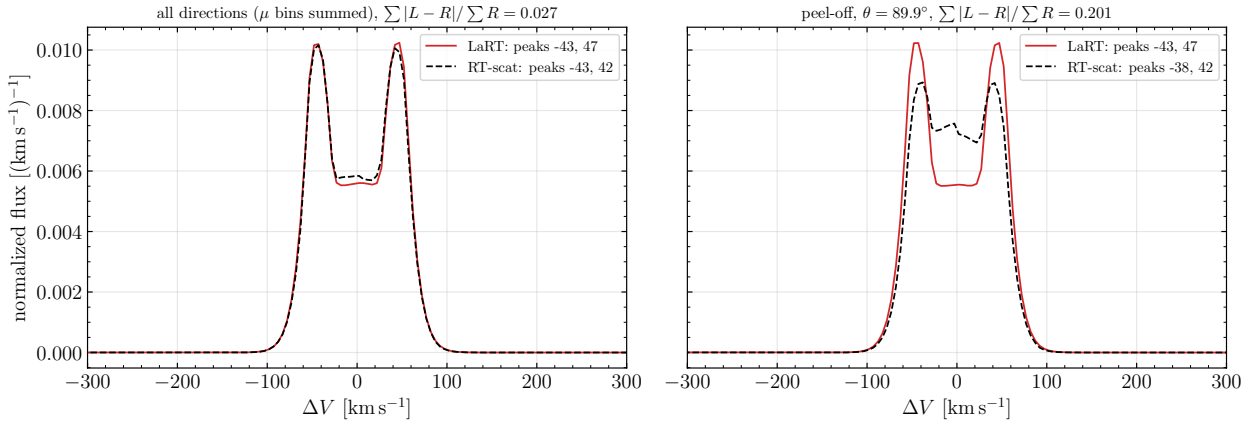


Figure 2: LaRT against RT-scat. Left: all directions (μ bins summed), $\sum|L-R|/\sum R = 0.027$. Right: peel-off at $\theta = 89.9°$, $\sum|L-R|/\sum R = 0.201$.

4.2 Empty center and overlapping clumps

RT-scat places no clumps at $r < 0.1R_o$ and does not prevent overlaps, whereas the paper (Section 3.2) describes uniformly distributed, non-overlapping clumps. The clump number density is that of the whole sphere, so the realized covering factor and mean radial column are $0.9f_c$ and $0.9N_{\rm HI}$.

4.3 Smooth sphere: grid sampling of the rotation field

The smooth rotating sphere is sampled on a 201^3 Cartesian grid. Inside $R_i = 0.1R_o$ the rotation velocity changes by $\Omega\Delta x = 30\,{\rm km\,s^{-1}}$ from one cell to the next, 2.3 times the thermal width. Seen from an azimuth that is not a mirror plane of the grid ($\phi = 60°$, used by RT-scat), this staircase makes the edge-on peel-off spectrum lopsided. The effect is a property of the grid, not of either code; it was measured with LaRT (smooth sphere, $N_{\rm HI} = 10^{18}\,{\rm cm^{-2}}$, $V_{\rm rot} = 300\,{\rm km\,s^{-1}}$, $\beta = 90°$, 10^6 photons; the AMR row has the fine structure off, the others on):

grid	step in R_i	red/blue flux, $\phi = 30°/60°$	red/blue peak, $\phi = 30°/60°$	$\phi = 0°/90°$ flux
101^3	59 km s^{-1}	0.936 / 1.071	1.054 / 0.958	1.007 / 0.998
201^3	30 km s^{-1}	0.983 / 1.021	1.056 / 0.953	1.001 / 1.001
401^3	15 km s^{-1}	1.001 / 1.003	1.012 / 0.988	1.006 / 0.995
AMR, levels 4–9	$\leq 1.29\,b$	0.997 / 0.999	0.999 / 0.983	1.001 / 0.995

At $\phi = 0°$ and $90°$ (mirror planes) the 201^3 spectrum is symmetric but rippled (Figure 3); $\phi = 30°$ and $60°$ have opposite signs, and $S_{120}(V) = S_{60}(-V)$ to 1.5%, as the $x = 0$ mirror symmetry of the grid requires. Switching the fine structure off changes nothing. The AMR grid (`make_amr_grid.py --vel_criterion doppler --vel_threshold 1.0 --level_max 9`, 1.1×10^7 leaves) reproduces 401^3 within 0.6–2.3% with 1/6 of the cells. The RT-scat smooth curve of the paper used 201^3 and $\phi = 60°$ and is therefore expected to carry this asymmetry.

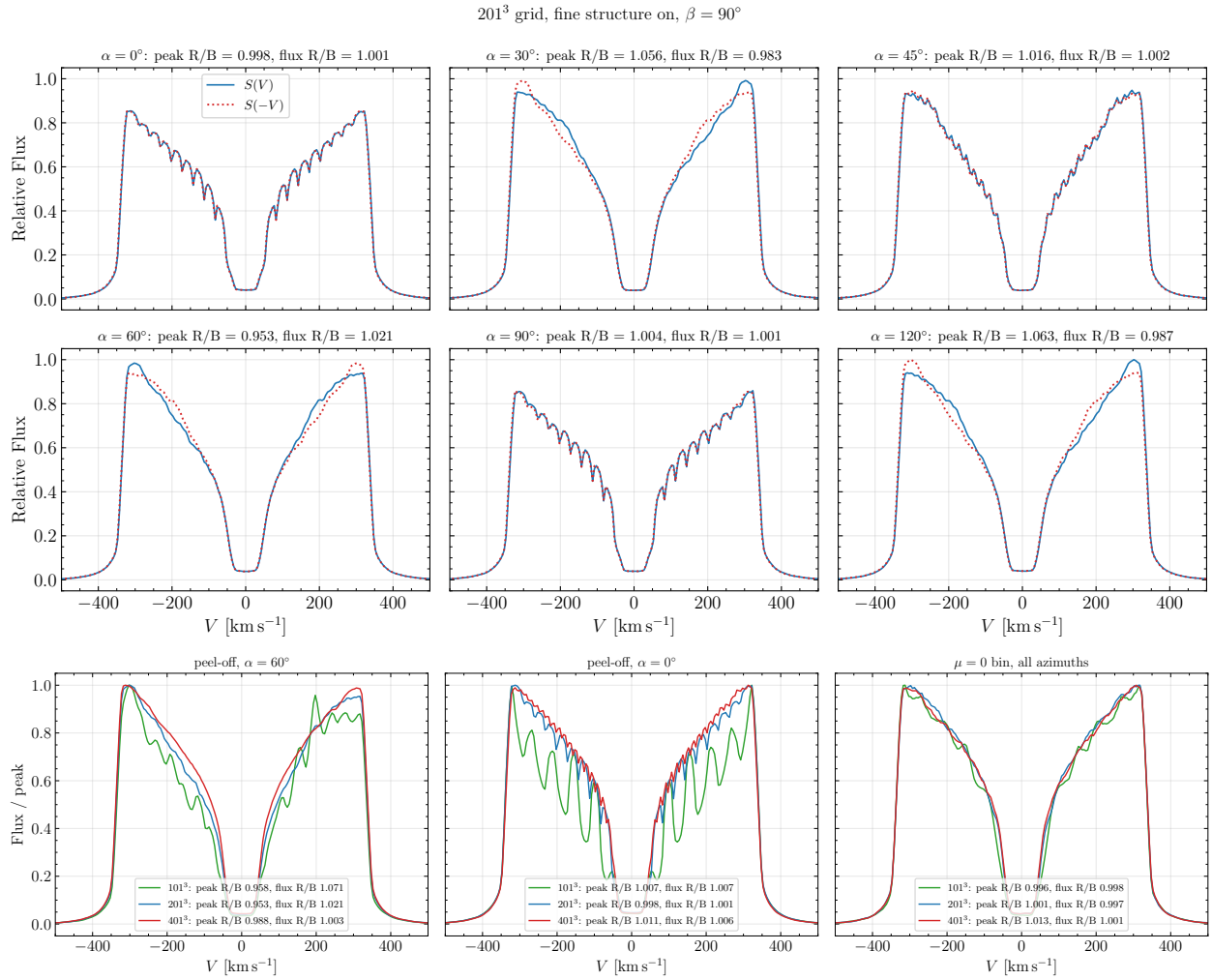


Figure 3: Smooth rotating sphere. Top: 201^3 peel-off spectra at six azimuths (solid) and their mirror images (dotted). Bottom: 101^3, 201^3, 401^3 at $\phi = 60°$, $\phi = 0°$ and in the $\mu = 0$ bin.

4.4 Direct light seen by a single observer

The unscattered light of the point source reaches a peel-off observer along one sight line. Moving that line by 0.1° shifts it at $r = 0.5R_o$ by $1.7 \times 10^{-3}R_o$, more than a clump radius, so for small f_c the direct spike can appear in one direction and vanish in the next. In one LaRT $f_c = 1$ realization (same photons, same clumps), the direct light was 59% of the flux at $\theta = 89.9°, \phi = 0°$ and zero at $\theta = 90°$; the scattered light differed by less than 1%.

4.5 Practical issues

Run parameters (column density, covering factor, rotation speed, photon number, observer angles) are set by editing the main program before compiling, so a run cannot be reproduced from its outputs alone. The build script compiles `*_mod.f90` in alphabetical order and fails unless the modules happen to be compiled after `cons_mod.f90`; repeating the compile command until all modules are built works. The observer angles 0.1° and 89.9° are not needed: the code already treats $|k_z| = 1$ separately and has no singularity at 90°.

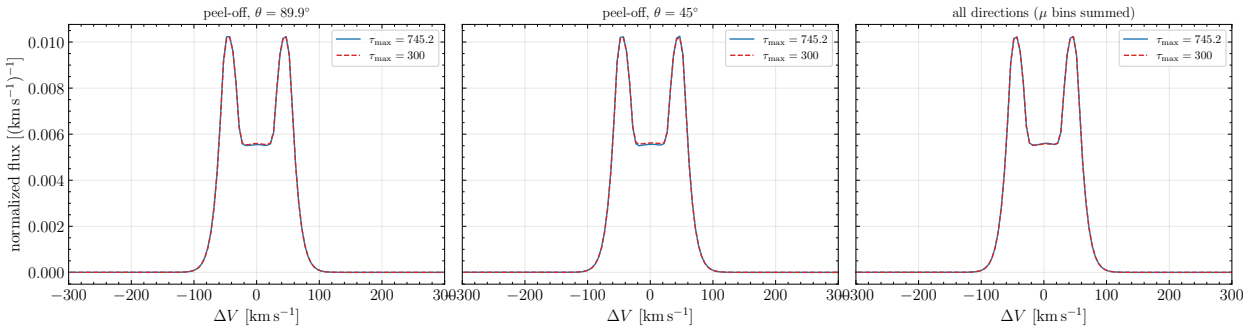


Figure 4: Static $f_c = 10$ spectra with the peel-off cutoff at $\tau = 745.2$ and $\tau = 300$ (different random realizations). Flux ratios 0.993–1.017 at the seven angles, 1.000 over all directions; shape differences 0.2–1.3%.

5. Changes made to LaRT

- **Peel-off τ cutoff.** The early exit of the photon-transport τ integrals used by the peel-off (Cartesian, AMR and clump) is now a single constant, `tau_huge = 300` in `define.f90`; it was 745.2 (where $e^{-\tau}$ underflows in double precision), defined separately in each routine. Beyond $\tau = 300$, $e^{-\tau} < 5 \times 10^{-131}$, so no peel-off weight changes; the static $f_c = 10$ run went from 76.3 to 42.2 min with the spectra unchanged (Figure 4). Sight-line τ outputs are integrated without the cutoff.

- **AMR refinement for velocity fields.** `make_amr_grid.py` and `AMRGrid` accept `vel_criterion='doppler'`: a leaf is refined while the bulk velocity changes across it by more than `vel_threshold` Doppler widths $b = \sqrt{2kT/m + b_{\rm turb}^2}$.

- **HDF5 format of the Python writers.** They write the HDF5 1.14 format (h5py `libver` upper bound `'v114'`); with h5py built on HDF5 2.0, `libver='latest'` produced files that LaRT (linked with HDF5 1.14) could not read.

- **`par%grid_type`** (`'car'`, `'amr'`, `'clump'`) is accepted by LaRT_v2.00 as in LaRT_v2.10; other values stop the run.

6. Open questions

- Why LaRT scatters twice as often as RT-scat in the static $f_c = 10$ model. Running LaRT with the RT-scat placement (empty center, overlaps) would separate the placement from the transport.

- Whether the published Figure 9 clumpy curves contain the sheet leak; this needs the RT-scat outputs used for the paper.